Computer Vision Assignment - Auric AI Technologies

Dataset	5-class labelled dataset with train and validation sets
Environment	Google Colab or equivalent
Tooling	Open internet; pretrained models, public code, papers and AI tools are allowed

Primary objective
Develop an object detector that achieves at least 0.75 mAP50 on the provided validation set. Reaching the target is necessary, but the quality of your diagnosis, experimental reasoning and reproducibility will be evaluated alongside the score.

Evidence standard
A statement such as “the model struggles on small objects” is not sufficient on its own. Show how you established the claim, how large the effect is, and what competing explanation could produce a similar observation.

1. Dataset and Task

You are provided with a five-class labelled dataset containing separate training and validation sets. Class definitions are supplied with the dataset.

Dataset link: [Dataset]

Model choice is intentionally open
You may choose the detector architecture, pretrained initialization, input resolution, augmentation policy, loss setup, sampling strategy and training schedule. Any external weights, code or tools must be documented clearly enough for us to reproduce the result.

2. Baseline, Dataset Observations and Reproducibility

1. Study the supplied dataset before or alongside model development. Report the observations that materially influence your modeling or evaluation decisions.
2. Establish a reasonable initial detection baseline and document the complete training setup required to reproduce it.
3. For every reported model, provide overall mAP50 and per-class AP50 on the validation set.
4. Include representative qualitative predictions and misclassification details. Do not limit the report to successful examples.

Minimum reproducibility information: architecture and initialization, image size/preprocessing, train/validation usage, augmentations, optimizer and scheduler, batch size, epochs, sampling strategy, losses or class weighting if modified, random seed where applicable, inference thresholds/post-processing, software dependencies and hardware/runtime details.

3. Diagnose What the Detector Is Really Getting Wrong

Do not treat every error reported by the evaluator as the same phenomenon. Diagnose the failures of your detection experiment and support the analysis with computed metrics and representative visual examples.

- Investigate false positives and false negatives as populations, not only as isolated examples.
- Develop a useful failure taxonomy and quantify the dominant failure modes.
- Report class-confusion / misclassification behavior and identify any systematic patterns.
- Where several explanations are plausible, state the alternatives and design an analysis or experiment that helps discriminate between them.

4. Experimentation Toward the Target

If your initial system does not achieve the stated objective, run additional experiments as needed. Each experiment should follow a traceable research chain rather than being an isolated hyperparameter trial.

Experiment Record

Field	What to record
Observation	The observation or evidence motivating the current experiment.
Hypothesis	Your explanation for the observation and the evidence that makes it plausible.
Changes vs. previous run	Exactly what changed relative to the initial model or previous experiment.
Results	Quantitative and qualitative results, including relevant class-wise behavior.
Conclusion	What the experiment supports, weakens or leaves unresolved.
Next experiment setup (if required)	How the conclusion informed the next experiment. State what uncertainty the next run is intended to resolve.

5. Research Investigations

Use the validation set for the investigations below. You may run additional controlled experiments where needed. Support conclusions with evidence and explicitly state the limits of what each experiment can establish.

5.1 If Object Locations Were Perfect

Suppose the location of every object were known perfectly. How much of the five-class problem would remain?

- Design and perform an experiment that answers this question.
- Quantify how much error remains once localization uncertainty is removed or controlled.
- Explain what the experiment still cannot establish.

5.2 Which Training Examples Resist Learning?

Aggregate training loss can hide very different behavior across individual examples.

- Find a principled way to identify training instances that the system repeatedly struggles to learn.
- Investigate whether these examples form one population or several distinct populations.
- Provide evidence for the mechanisms you believe are responsible. “Hard examples” by itself is not an explanation.
- Where possible, use the observed training behavior to make a prediction about examples you have not manually inspected, then test that prediction.

5.3 Would Another 500 Labels Matter?

> **Research expectation**
> The purpose of the reduced-data investigation is not only to compress the dataset. Use it to learn something about redundancy, coverage, class balance and the effective value of individual examples.

Assume 500 additional labelled instances can be collected from the same broad data source and labeling process as the current training set.

- Estimate whether 500 additional labels would materially improve the system.
- State the assumptions behind your estimate and the uncertainty in the conclusion.
- Identify which of the five classes are likely to benefit most and least, and support the ranking with evidence from your experiments.
- Distinguish, as far as the available evidence permits, between limited sample count and other possible performance limitations.

5.4 Find a Small Effective Training Subset

Goal: Find the smallest training subset you can demonstrate recovers at least 90% of the full-data validation mAP50.

- Declare the subset-selection method before training the reduced-data model.
- Compare the selected subset against a random subset of the same size under a comparable training setup.
- Characterize the examples that survive selection. What makes them useful?
- Report subset size, class coverage and achieved performance.
- Report the smallest successful subset you actually tested. Do not claim a mathematically proven minimum unless you can establish one.

6. Final Analysis

Your final report should connect the experiments into a coherent argument. Avoid presenting a sequence of unrelated runs.

- What are the dominant limitations of the final detector?
- Which conclusions are strongly supported, which remain plausible but unresolved, and which initial hypotheses were weakened or rejected?
- What changed most between the initial and final system, and what evidence supports your explanation for that change?
- What is the single highest-priority next experiment or data action you would take if given one additional working day? Explain why it has higher information or performance value than the alternatives.

7. What to Submit

Deliverable	Expectation
Concise technical report	Dataset observations, baseline, experiment records, quantitative and qualitative results, failure analysis, research investigations, conclusions and prioritized next step.
Final model weights	The exact checkpoint corresponding to your claimed validation result.
Inference and evaluation code	A reproducible path from provided validation images to predictions and reported metrics.
Training code/configuration	All configurations, scripts and commands required to reproduce the final training setup.
Environment/dependencies	Requirements file, environment specification or equivalent, including key package versions.
Run instructions	Short README with commands and expected inputs/outputs.

Submission standard A reviewer should be able to reproduce your claimed validation predictions and metrics without reconstructing undocumented choices. External resources and AI-assisted code are permitted, but you remain responsible for understanding, documenting and defending the submitted system and conclusions.